

DARE: Differential Adversarial Reward Equalization for Humanoid Motion Imitation

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Abstract—Physics-based humanoid motion imitation must simultaneously satisfy several heterogeneous objectives, including global root motion, articulated pose, and velocities, keeping them consistent with a synchronized reference trajectory. Handcrafted rewards combine these objectives with manually designed weights and scales, whereas adversarial imitation can learn a unified imitation signal from data through a discriminator, reducing the reliance on explicit reward weights. For differential discriminators that take the reference-policy differential as input, the sensitivity of the score function and its numerical range still have to be controlled; otherwise, changes in the residual distribution across motions can make the same regularization setting produce different effective learning signals. We propose DARE, a differential adversarial reward method that requires no motion-specific coefficient and controls the geometry of the score function and the score scale separately. DARE maps each semantic residual block through its own map into a representation of a common latent width, with an independent spectrally normalized gain bound imposed on each map, thereby bounding the score change relative to the unique zero-error anchor. DARE then performs a one-shot closed-form score calibration according to the response window of the reward transform, with the global scaling factor determined automatically from the residual distribution observed during training. All motions use the same network architecture, reward definition, and calibration rule, without re-selecting reward or regularization coefficients for different motions. We evaluate DARE with a single configuration on six humanoid skills with diverse dynamical characteristics and contact patterns. Experimental results show that DARE attains the lowest cross-motion macro-average error on all four geometric tracking measures, together with the foot progress closest to the reference and the lowest lateral drift, and we further deploy the resulting controllers on **[TODO: robot platform]**.

I. INTRODUCTION

Physics-based humanoid motion imitation requires the controller to reproduce a prescribed reference motion under dynamic constraints, actuator limits, and complex contact conditions. Whether an imitation succeeds is not a single geometric question. The root must translate and turn as the reference does, the articulated body must maintain consistent poses, joint and root velocities must stay synchronized with the reference motion, and contacts must occur at the right moments. Because these objectives have different physical units, numerical ranges, and dynamical characteristics, and are coupled through the robot dynamics, motion imitation is inherently a multi-objective control problem.

Traditional methods design a separate reward term for each type of tracking error and combine them into a scalar reward through manually chosen weights and scales. A

representative form is

$$r_t^{\text{hand}} = \sum_{g=1}^G w_g \exp\left[-\alpha_g \|\phi_g(\hat{s}_t) - \phi_g(s_t)\|_2^2\right], \quad (1)$$

where ϕ_g extracts the g -th motion feature—joint rotations, root position, velocities, or end-effector positions—and s_t and $\hat{s}_t$ denote the current simulated state and the synchronized reference state. The weights w_g decide the relative importance among tracking objectives, and the scales α_g fix the numerical range of each error in the reward. The difficulty is that typical error magnitudes, dynamical difficulty, and the relative importance of the objectives change with the motion, so a coefficient set that serves one skill need not serve another; in practice, reward design is re-tuned whenever the motion changes.

Adversarial imitation offers a way to reduce manual reward design. Instead of declaring in advance how many meters of root-position error are worth how many radians of rotation error, a discriminator learns a unified imitation signal directly from data. AMP and related methods learn rewards that describe motion style and plausibility by discriminating whether short state transitions come from the reference distribution [7]. For motion tracking that must strictly follow a synchronized reference trajectory, ADD further feeds the reference-policy state differential to the discriminator and asks whether it is close to the ideal zero differential [8]. The aggregation among tracking objectives can then be learned by the discriminator, instead of specifying the per-term reward weights and scales of (1) by hand.

Differential adversarial rewards, however, still face an important issue. Their ideal sample is always a single point, the zero differential. Without a suitable constraint, the discriminator easily becomes too sharp: it assigns a high score to the origin and compresses the many nonzero residuals produced by the policy into a nearby low-score region. Even if classification succeeds, the discriminator may not provide an effective continuous learning signal for the policy. Existing methods constrain the discriminator through regularization, but one regularizer simultaneously affects two distinct properties. On the one hand, it controls the discriminator’s local sensitivity, that is, how fast the score changes with the input residual. On the other hand, it also affects the numerical range in which the final scores live, and this range in turn determines whether the policy reward operates in an effective interval. As the residual distribution changes across motions, the same regularization setting can correspond to different sensitivities and different score scales, producing

motion-dependent effective regularization requirements.

The core idea of DARE is to treat these two issues separately rather than control them with one motion-dependent coefficient. Consider first the critic’s sensitivity to different tracking objectives. For any motion and any phase, perfect imitation always corresponds to the same zero differential $\Delta = 0$, which we call the anchor. Meanwhile, the differential in motion tracking naturally decomposes into semantic blocks with clear physical meaning, such as root motion, body pose, and velocities. Existing differential discriminators feed these raw residual coordinates directly into one network, but the feature dimensions occupied by different objectives can differ greatly. A shared first layer may then tie the critic’s sensitivity to an objective to its representation dimension rather than to the learning value of the objective itself.

DARE therefore processes the semantic residual blocks separately before handing them to a shared critic. Each block enters a common-width latent representation space through its own spectrally normalized map, providing every objective with a structurally consistent and mutually independent input interface. The shared critic then learns the relations among objectives from the residuals that actually occur during training. This design does not force equal importance across objectives; it prevents them from receiving different structural amplification before entering the critic merely because of unequal representation widths. Spectral normalization further bounds the maximum rate at which the score changes with the residual, so that the same critic structure can be used stably across motions. The concrete construction and the corresponding Lipschitz properties are given in Section IV-B.

Constraining the critic’s sensitivity alone does not resolve the score scale. A Lipschitz bound states how fast the score may at most change, not how large a numerical range the trained critic actually uses. Different motions induce different residual distributions, so even with identical network structures and constraints the resulting score ranges can still differ. These scores further pass through a nonlinear reward transform before becoming the learning signal. If the score range is too large, most policy states may fall in a region where the reward changes very little, so improving poor motion earns almost no additional reward. If the range is too small, the reward differences between tracking qualities are compressed, leaving the reward with insufficient discriminative power.

DARE therefore performs one automatic score calibration after the input normalization stabilizes. It measures the score gap between the zero-error anchor and a typical policy residual, and rescales the entire critic score uniformly according to the response range of the reward transform itself. All motions use exactly the same calibration rule, and no new motion-specific coefficient is chosen during calibration; the only quantity that changes across motions is the residual statistic measured automatically from the training data. In other words, DARE uses the network architecture to control how the critic responds to different types of residuals, and one data-driven closed-form calibration to control the operating scale at which these scores enter the reward function. The

critic’s geometry and score scale thus no longer depend on the same hand-tuned parameter.

The main contributions are threefold. First, we identify that the regularizer in existing differential adversarial rewards simultaneously affects the critic’s local geometry and its score scale, and that this coupling can make different motions require different effective regularization settings. Second, we propose DARE, which handles the semantic residual blocks separately through a factorized spectrally normalized critic and sets the score scale automatically by a one-shot anchor calibration, removing the need to re-specify reward or regularization coefficients for each motion. Third, we evaluate one DARE configuration on six skills with different motion durations, contact patterns, and dynamical characteristics, and deploy the resulting controllers on [\[TODO: robot platform\]](#).

II. RELATED WORK

Physics-based motion tracking and learned imitation objectives. Physics-based character control has progressed from skill-specific, carefully engineered tracking objectives to learning-based systems that cover larger motion collections. DeepMimic combines pose, velocity, root, and end-effector tracking terms for physics-based skills [1]; subsequent work improves scalability and motion coverage through architectures, representations, and data [2]–[5]. Adversarial imitation provides a learned alternative to a manually written reward: GAIL matches expert and policy occupancies [6], while AMP learns a motion prior by discriminating short state transitions in physics-based characters [7]. These distribution-level objectives encourage plausible behavior, whereas reference-conditioned tracking requires agreement with a synchronized trajectory. ADD addresses this setting by presenting the reference–policy differential to an adversarial discriminator and learning a nonlinear aggregation of tracking objectives without component-wise reward weights [8].

Multi-objective learning and objective aggregation. Competing objectives are commonly combined by weighted scalarization, Pareto optimization, gradient balancing, adaptive loss weighting, or multiple objective critics [14]–[16]. These approaches differ in whether preferences are specified explicitly, adapted from objective statistics, or retained as separate signals. Adversarial differential learning offers another form of scalarization in which the aggregation is represented by a learned nonlinear function [8]. Motion tracking adds a representation issue to this problem: one physical objective can occupy a feature block with many more coordinates than another, even when both are conceptually single objectives.

Regularization of adversarial critics. Adversarial critics are regularized with gradient penalties, Lipschitz constraints, or spectral normalization to control sensitivity and improve optimization [9]–[11]. Gradient penalties constrain sampled input derivatives, whereas spectral normalization bounds the gain of individual network maps by construction. For non-Wasserstein adversarial losses, the numerical score range also determines the operating regime of the loss and its reward

transform [12], [13]. Geometry and score calibration are therefore related but distinct considerations when a critic supplies the learning signal.

III. BACKGROUND

A. Reference-Conditioned Motion Tracking

In reference-conditioned humanoid motion imitation, the controller must keep the executed motion continuously aligned with a time-synchronized reference trajectory. Let s_t and $\hat{s}_t$ denote the current state and the reference state at time t . Tracking quality is jointly measured by several physical quantities, including root position and orientation, articulated body pose, and motion velocities. We organize these quantities into G semantic feature groups: $\phi_g(s_t)$ extracts the feature vector of group g from the current state, and $\phi_g(\hat{s}_t)$ its counterpart from the reference state. Different groups may have different dimensions and physical units—positions are measured in meters, while rotations and velocities have their own representations. Motion tracking is therefore not the minimization of a single error, but the simultaneous reduction of multiple tracking errors with different representations and scales.

B. Handcrafted Objective Scalarization

Traditional methods first compress each tracking error into a scalar. For group g , the scalar tracking error at time t is the Euclidean distance $e_{g,t} = \|\phi_g(\hat{s}_t) - \phi_g(s_t)\|_2$, which is zero when the group matches the reference exactly and grows with the corresponding deviation. The errors $e_{g,t}$ are then transformed and combined into a single policy reward. DeepMimic is a representative instance of this class: its tracking terms use weights and error scales specified before training and held fixed throughout [1]. This procedure is in essence an objective scalarization. The original problem contains multiple tracking objectives while policy optimization consumes a scalar reward, so a trade-off among the errors must be fixed; handcrafted rewards fix this relation through constant coefficients chosen before training starts. Because typical error magnitudes and dynamical characteristics differ across motions, one coefficient set need not serve all motions. Learned scalarization instead leaves the trade-off unspecified in advance and learns the aggregation among objectives from the error distribution that actually occurs during training.

C. Differential Adversarial Objectives

Differential adversarial learning provides such a learned aggregation [8]. Consider n non-negative objective losses $l^1, \dots, l^n$ collected into a differential vector $\Delta = (l^1, \dots, l^n)$, each entry recording the gap between the current solution and the ideal on one objective; when all objectives are simultaneously ideal, $\Delta = 0$. A discriminator $D(\Delta)$ is trained to separate the ideal zero differential from the nonzero differentials produced by the current policy. Since the discriminator receives the differentials of all objectives at once, it can learn a nonlinear objective aggregation function without fixed per-objective weights. Its local sensitivity to objective i is described by $\partial D / \partial l^i$: the extent to which the discriminator

output responds when objective i changes slightly near the current differential. Unlike fixed scalarization weights, this local sensitivity can vary with the position in the residual space and during training. In synchronized motion tracking, the differential is no longer restricted to one scalar loss per objective but can be formed directly from the feature residual between the current and reference states. Perfect tracking again corresponds to the unique zero differential, while nonzero differentials retain the error direction and the relations among features. The policy reward is generated from the discriminator's score on the current differential, for example $r_t = -\log(1 - D(\Delta_t))$: the closer the current residual is judged to the ideal state, the higher the reward.

A peculiarity of this form is that the ideal class always concentrates on the single point $\Delta = 0$. If the discriminator is overly sensitive to input changes, it can easily separate the origin from nonzero residuals, yet fail to form a smooth and useful score among policy residuals of different quality. Differential adversarial methods therefore constrain the local sensitivity of the score function, for example with a sampled gradient penalty $\lambda \mathbb{E}_{\Delta \sim q_\pi} \|\nabla_\Delta D(\Delta)\|_2^2$, where q_π denotes the residual distribution produced by the current policy [8]. Note that constraining the local sensitivity does not fully determine the score range the trained discriminator uses: the former describes how fast the score changes with the residual, the latter describes where the trained scores are distributed numerically. Since the policy reward further passes through a nonlinear transform, both properties affect the final learning signal, but they are not the same question.

IV. DARE: DIFFERENTIAL ADVERSARIAL REWARD EQUALIZATION

A. Learned Differential Reward

Motion tracking involves multiple objectives with different physical meanings. For each objective, the scalar distance $e_{g,t}$ summarizes the error magnitude between the current and reference states and is well suited to a traditional weighted reward. A scalar distance, however, retains only the magnitude of the error. The raw feature differential also contains the error direction and the joint variation across coordinates, and this information likewise reflects how the current motion departs from the reference trajectory.

DARE takes these feature differentials directly as the input of the learned reward. For semantic group g , the raw residual between the current state and the synchronized reference state is $\phi_g(\hat{s}_t) - \phi_g(s_t)$. Different groups have different physical units and typical numerical ranges. If positions, rotations, and velocities were concatenated directly into one vector, their numerical scales would influence the input seen by the subsequent network. We therefore normalize every coordinate first. Let the normalizer of group g be $N_g = \text{diag}(\nu_g)^{-1}$, where ν_g records the mean absolute magnitude of each coordinate's residual. These statistics are estimated from residual samples in early training and fixed after the first 10^8 training samples, as is standard for adversarial motion rewards. The normalization only rescales coordinates and introduces no offset, so the zero differential corresponding

to perfect tracking always stays at the origin; rotations use a tangent-normal representation so that the corresponding residuals can be handled uniformly in Euclidean coordinates.

After normalization, the group residual, the full differential, and the ideal point are

$$\Delta_{g,t} = \mathcal{N}_g(\phi_g(\hat{s}_t) - \phi_g(s_t)), \quad \Delta_t = \bigoplus_{g=1}^G \Delta_{g,t}, \quad \Delta^+ = 0. \quad (2)$$

The vector $\Delta_t \in \mathbb{R}^d$, $d = \sum_g d_g$, retains the direction of each coordinate error and the structural relations formed when several coordinates vary together. Whatever the reference motion and whatever the current phase, perfectly consistent states map to the same zero differential, while the nonzero differentials produced by the policy record how the current motion deviates from the reference. During training, the current policy keeps producing new differential samples, and recently visited differentials are kept in a replay buffer; the two parts together form the policy residual distribution q_π . As the policy gradually learns the reference motion, q_π keeps evolving with training, so the critic always sees the residual region that the current controller actually needs to improve.

DARE learns a scalar score function on this differential space. Let $f_\theta(\Delta)$ denote the raw scalar score of the critic, converted by the logistic sigmoid into $D_\theta(\Delta) = \sigma(f_\theta(\Delta))$, which expresses the degree to which the critic considers the differential ideal. The zero differential represents perfect tracking, and the residuals produced by the policy are drawn from q_π ; the critic is therefore trained by the following adversarial objective to distinguish the two:

$$\max_{\theta} \log D_\theta(0) + \mathbb{E}_{\Delta \sim q_\pi} [\log(1 - D_\theta(\Delta))]. \quad (3)$$

The first term raises the ideality score of the zero differential, and the second lowers the ideality score of current policy residuals. As the policy improves, the residual distribution gradually moves toward the origin, and the critic updates its score function accordingly. A function f_θ that co-evolves with training thereby aggregates multiple tracking objectives into a single scalar signal.

For optimization, we write the above binary classification objective as a balanced Bernoulli negative log-likelihood. Using $-\log \sigma(z) = \text{softplus}(-z)$ and $-\log(1 - \sigma(z)) = \text{softplus}(z)$ gives the critic loss

$$\mathcal{L}_{\text{DARE}}(\theta) = \frac{1}{2} \text{softplus}[-f_\theta(0)] + \frac{1}{2} \mathbb{E}_{\Delta \sim q_\pi} \text{softplus}[f_\theta(\Delta)]. \quad (4)$$

This objective provides the critic with only two kinds of supervision, ideal differentials and policy residuals. How root motion, body pose, and velocities jointly determine the final score is learned by f_θ from the residual distribution observed during training, and the critic's response to each type of residual adjusts as q_π changes.

The policy then constructs its reward directly from the critic score. DARE defines

$$r_t^{\text{DARE}} = -\beta \log[1 - D_\theta(\Delta_t)] = \beta \text{softplus}[f_\theta(\Delta_t)], \quad (5)$$

where β controls the overall reward amplitude and is held fixed across all motions. The closer the critic judges the

current differential to the ideal state, the higher $f_\theta(\Delta_t)$ and the larger the policy reward. The policy can thus improve root motion, body pose, and velocities simultaneously through one unified scalar reward.

This formulation recasts multi-objective tracking as learning a scalar score function in the differential space, and its effect depends on two properties of the critic itself. First, the semantic feature groups in the differential have different dimensions, and the critic's sensitivity to each group's residual variation requires a clear and consistent structural constraint. Second, the scores produced by the critic are converted into the policy reward through the softplus transform, so the numerical range of the scores directly determines how strongly the reward responds to changes in tracking quality. DARE states these two properties as critic geometry and score scale, and designs for them independently.

B. Equalized Critic: Geometry by Construction

The full differential Δ consists of G semantic feature blocks, the g -th block denoted $\Delta_g \in \mathbb{R}^{d_g}$, where d_g is the feature dimension of that objective. Different objectives in motion tracking use different physical representations, so their coordinate counts usually differ. Per-coordinate normalization makes the typical numerical scales of coordinates comparable, but the overall norm of a feature block is still influenced by its dimension. When coordinates have comparable typical magnitudes, the block norm $\|\Delta_g\|_2$ generally grows with $\sqrt{d_g}$. Consequently, if all semantic blocks were concatenated and fed directly into one shared dense layer, the raw representation dimension would participate in deciding how strongly the network responds to each objective's residual variation.

Let this shared linear map be $W = [W_1 \cdots W_G]$, where W_g acts on the g -th semantic block. The largest singular value $\sigma_{\max}(W)$ of a linear map expresses the maximum amplification it can apply to an input change, and imposing $\sigma_{\max}(W) \leq 1$ means that distances between any two points are at most preserved—a 1-Lipschitz property. For any unit output direction u , the shared map satisfies

$$\sum_{g=1}^G \|W_g^\top u\|_2^2 = \|W^\top u\|_2^2 \leq 1, \quad (6)$$

which shows that the semantic blocks are jointly subject to one overall gain constraint in each output direction. A shared layer can bound the maximum amplification of the whole map, but the gains of the blocks are allocated by the same matrix, so its response is coupled to the input representation itself. For semantic blocks of different dimensions, this coupling lets the representation dimension influence how sensitive the critic is to each objective's residual variation.

DARE instead builds an independent constrained representation for each semantic block before the blocks interact. The g -th block is mapped into the latent space by

$$\psi_g(\Delta_g) = \rho(\bar{W}_g \Delta_g + b_g), \quad \bar{W}_g = \frac{W_g}{\sigma_{\max}(W_g)}, \quad (7)$$

where ρ is a 1-Lipschitz activation. Spectral normalization rescales the largest singular value of each linear map to one, so each ψ_g carries its own independent 1-Lipschitz gain bound: $\|\psi_g(\Delta_g) - \psi_g(\Delta'_g)\|_2 \leq \|\Delta_g - \Delta'_g\|_2$. That is, a residual change within one semantic block is never amplified beyond its original magnitude after passing through its own map.

All semantic blocks use the same latent width m , so $W_g \in \mathbb{R}^{m \times d_g}$ and $\psi_g(\Delta_g) \in \mathbb{R}^m$. The original feature dimensions d_g may differ, but before entering the shared critic every objective is represented by an m -dimensional latent vector. This uniform width assigns each semantic objective the same number of representation channels in the subsequent shared network while preserving its own independent input gain bound. When $m = 1$, a block is compressed into a single latent scalar; when $m > 1$, the block is represented jointly by several latent coordinates and can therefore retain more within-block residual structure. We adopt the vector representation so that each objective forms its own latent features before interacting with the others.

The latent representations of the blocks are then combined by a direct sum, $\Phi_{\oplus}(\Delta) = \bigoplus_{g=1}^G \psi_g(\Delta_g)$, where $\bigoplus$ places each m -dimensional latent vector in mutually disjoint coordinates; with G semantic blocks the total dimension of $\Phi_{\oplus}(\Delta)$ is Gm . Within-block features are thus processed first by their own maps, and cross-objective information interaction happens uniformly in the subsequent shared trunk. The shared trunk aggregates these latent representations into a scalar score, $g_{\theta}(\Delta) = \tilde{T}(\Phi_{\oplus}(\Delta))$ with $\tilde{T}(h) = \bar{w}^T \rho(\bar{V}h + c) + b$, whose linear maps are spectrally normalized in the same way so that $\text{Lip}(\tilde{T}) \leq 1$. Here g_{θ} denotes the raw critic output before calibration; the score f_{θ} used in the loss and the policy reward is identical to g_{θ} before the scale calibration and afterwards differs by one fixed global scaling factor. All affine layers keep their bias terms, so the activation boundaries can adjust to where the residual distribution actually lies. Biases shift the network’s response regions in input space, while spectral normalization continues to bound how much an input change can change the output; the two play the roles of position adjustment and gain constraint, respectively.

The overall property of this block structure follows directly from a distance relation. Because the latent representations of the semantic blocks occupy non-overlapping coordinates, for any two differentials Δ and Δ' ,

$$\begin{aligned} \|\Phi_{\oplus}(\Delta) - \Phi_{\oplus}(\Delta')\|_2^2 &= \sum_{g=1}^G \|\psi_g(\Delta_g) - \psi_g(\Delta'_g)\|_2^2 \\ &\leq \sum_{g=1}^G \|\Delta_g - \Delta'_g\|_2^2 = \|\Delta - \Delta'\|_2^2, \end{aligned} \quad (8)$$

so the entire map from the raw differential to the concatenated latent representation remains 1-Lipschitz. Each semantic block first passes through its own independent constrained map, and the shared trunk then learns the relations among objectives. The uniform latent width specifies a consistent representation interface, the independent spectral

normalization specifies each block’s own gain bound, and how sensitive the critic should actually be to each objective is learned by the shared trunk from the residual distribution observed during training.

Combining the non-expansive latent maps with the Lipschitz bound of the shared trunk yields, for the raw critic, $|g_{\theta}(\Delta) - g_{\theta}(\Delta')| \leq \|\Delta - \Delta'\|_2$. In motion tracking, perfect imitation always corresponds to the zero differential $\Delta = 0$, the anchor; taking $\Delta' = 0$ gives

$$|g_{\theta}(0) - g_{\theta}(\Delta)| \leq \|\Delta\|_2. \quad (9)$$

This relation directly constrains how fast the critic score can change relative to the anchor. When the current motion produces only small normalized residuals, the critic score can vary only within the corresponding range; larger score differences become possible only as the residual grows. The bias terms cancel in the difference of two scores, so they can adjust activation positions while keeping this bound unchanged. Together these constraints determine the critic’s geometry in the differential space. Here geometry describes how fast the score may change when the input residual changes, including the local sensitivity and the global Lipschitz bound of the whole score function. Through the independent semantic block maps, the uniform latent width, and the constrained shared trunk, DARE makes this property a direct consequence of the network structure.

C. Anchor Calibration: Scale in Closed Form

Spectral normalization limits the maximum rate at which the critic score may change with the residual. By the geometric constraint of the previous subsection, $|g_{\theta}(0) - g_{\theta}(\Delta)| \leq \|\Delta\|_2$, so the critic cannot produce arbitrarily large score changes for small residual perturbations. This bound, however, only specifies how fast the score function may at most change; how large a numerical range the trained critic actually uses is still decided jointly by the optimization process and the residual distribution.

The score range directly affects the reward received by the policy. DARE converts the critic score into the policy reward through $r^{\text{DARE}} = \beta \text{softplus}(f_{\theta})$, so the rate of change of the reward with respect to the score is

$$\frac{\partial r^{\text{DARE}}}{\partial f_{\theta}} = \beta \sigma(f_{\theta}), \quad (10)$$

which determines how much reward change a given score change can produce. When f_{θ} is very negative, $\sigma(f_{\theta})$ is close to zero, and score changes bring only tiny reward differences. As f_{θ} passes through the transition region around zero, the reward starts to respond markedly to score changes. When f_{θ} is large, the slope gradually approaches β , and softplus enters a nearly linear regime. The critic therefore needs not only a suitable sensitivity to residuals but also scores that fall inside the numerical range where the reward transform is sufficiently discriminative. The geometric constraint controls how residual changes become score changes, whereas the score scale controls where these scores operate after the reward transform.

This scale can be read from the adversarial loss itself. Let the anchor score be $f_\theta(0)$. The residuals produced by the policy follow q_π , and the critic’s mean ideality probability over these residuals is $\mathbb{E}_{\Delta \sim q_\pi} \sigma(f_\theta(\Delta))$. Mapping this mean probability back to the logit space defines the representative score of the typical policy residual, $\bar{f}_\theta = \sigma^{-1}(\mathbb{E}_{\Delta \sim q_\pi} \sigma(f_\theta(\Delta)))$. The balanced Bernoulli adversarial loss is stationary with respect to a global output shift when $\mathbb{E}_{q_\pi} \sigma(f_\theta(\Delta)) = \sigma(-f_\theta(0))$, hence $\bar{f}_\theta = -f_\theta(0)$. Writing $M = f_\theta(0) - \bar{f}_\theta$ for the score gap between the anchor and the typical residual then gives

$$f_\theta(0) = +\frac{M}{2}, \quad \bar{f}_\theta = -\frac{M}{2}. \quad (11)$$

At the reference placement induced by the balanced loss, the anchor score and the typical policy residual score are symmetric about zero, and M measures how far apart they are separated in logit space.

This gap directly determines the operating regime of the reward signal. When M is large, the typical policy residual sits in a deeply negative score region. The typical residual score $\bar{f}_\theta = -M/2$ lies far from the main sigmoid transition region and $\sigma(\bar{f}_\theta)$ is close to zero, so the reward and the reward slope at typical residuals are both small; even a meaningful improvement of a poor residual produces only a limited reward change. The critic can still distinguish the anchor from residuals well, but this separation already lies outside the range where the policy reward is most useful. When M is small, the anchor and the typical residual both concentrate near zero. Their reward-slope ratio $\sigma(M/2)/\sigma(-M/2) = e^{M/2}$ approaches one as $M \rightarrow 0$, while $r^{\text{DARE}} \approx \beta \log 2$ becomes the dominant reward level shared by most states; the score gaps between different tracking qualities are small, and the corresponding reward differences are compressed after the transform. A suitable score scale should therefore let the anchor and the typical policy residual span the main transition region of the sigmoid, so that changes in tracking quality translate stably into reward differences.

A sigmoid has no strict transition boundary, so the operating region needs an operational definition of its width. A common choice is the span over which the sigmoid climbs from 20% to 80% of its maximum. For the slope $\beta\sigma(f_\theta)$ of the DARE reward, the corresponding logit width is

$$w = \sigma^{-1}(0.8) - \sigma^{-1}(0.2) = \ln 4 - (-\ln 4) = \ln 16 \approx 2.77. \quad (12)$$

DARE adopts $w = \ln 16$ as the critic’s target operating gap. Under the symmetric reference placement, this width corresponds to $+w/2$ and $-w/2$, whose sigmoid values are 0.8 and 0.2, respectively. At this setting the critic already distinguishes the ideal state from typical residuals clearly, while both ends remain inside the main response region of the sigmoid. This operating region is fixed by the reward transform before training and is identical across all motions.

The critic obtained in actual training need not naturally use this scale. DARE therefore measures the score gap M_f of the current critic once, after the input normalization stabilizes,

and rescales the whole logit uniformly by

$$f_\theta(\Delta) = \kappa g_\theta(\Delta), \quad \kappa = \frac{w}{M_f},$$

$$M_f = g_\theta(0) - \sigma^{-1}(\mathbb{E}_{q_\pi} \sigma(g_\theta(\Delta))) \Big|_{t=t_f}, \quad (13)$$

where g_θ is the raw critic output before calibration and f_θ the calibrated score that enters the adversarial loss and the policy reward.

This scaling rule covers both kinds of scale deviation. When the measured gap M_f exceeds the target width w , $\kappa = w/M_f < 1$ and all critic scores are compressed toward zero, bringing an overly wide score gap back into the main response range of the reward transform. Conversely, when $M_f < w$, $\kappa > 1$ and the critic scores expand away from zero, enlarging the reward differences among tracking qualities. For instance, a measured $M_f = 2w$ gives $\kappa = 1/2$, halving the critic’s logit range, whereas $M_f = w/2$ gives $\kappa = 2$, doubling the score differences. These examples illustrate the scale action of the calibration. Because the typical residual score used in practice is defined through $\sigma^{-1}(\mathbb{E}\sigma(\cdot))$, the goal of the calibration is to move the overall logit scale toward the operating range set by w ; it does not require every post-calibration statistic to land exactly on a prescribed endpoint.

The gap M_f is measured once at the iteration t_f at which the input normalizer $\mathcal{N}$ is fixed. The computation uses the mixture of current-rollout and replayed residuals, consistent with the non-ideal sample distribution used in critic training. Before t_f , $\kappa = 1$: only after the normalizer is fixed do the units and scales of the differential coordinates stop changing, so a measured score gap first acquires a stable numerical meaning then. After the calibration, κ is held fixed, and both the critic loss and the policy reward use $f_\theta = \kappa g_\theta$; the critic continues to train at one scale, and the policy reads its reward at the same scale.

The uniform rescaling preserves the critic’s original score ordering and multiplies the raw score function by the same constant. The pre-calibration geometric bound $|g_\theta(0) - g_\theta(\Delta)| \leq \|\Delta\|_2$ accordingly becomes $|f_\theta(0) - f_\theta(\Delta)| \leq \kappa \|\Delta\|_2$. In this way, the network structure first fixes the geometric constraint on the scores, and the actual residual distribution then determines the global score scale. The target width $w = \ln 16$ is fixed by the reward transform, and M_f is measured automatically from the training data, so all motions share the same calibration rule while the resulting κ varies automatically with each motion’s residual distribution.

D. Training

Training alternates rollout collection, differential construction, critic updates, and PPO updates (Algorithm 1). Critic minibatches draw an equal mixture of current and replay differentials, with the ideal point presented once per minibatch as the positive class; at t_f , before the rewards of that iteration are computed, M_f is measured with the critic in evaluation mode and κ is set by (13). One critic, one reward, one policy, and one calibration rule make up the

Algorithm 1 DARE Training

Require: reference motion $\mathcal{M}$, policy π_φ , value V_ω , raw critic g_θ , replay buffer $\mathcal{B}$, normalizer $\mathcal{N}$; $\kappa \leftarrow 1$

- 1: **for** iteration $t = 1, 2, \dots$ **do**
- 2: Collect a synchronized rollout with π_φ from random reference phases; construct Δ_t by (2); add to $\mathcal{B}$
- 3: **if** $\mathcal{N}$ was fixed this iteration **then**
- 4: $\bar{g} \leftarrow \sigma^{-1}(\mathbb{E}_{\mathcal{B}_{\text{Urollout}}} \sigma(g_\theta(\Delta)))$
- 5: $M_f \leftarrow g_\theta(0) - \bar{g}$; $\kappa \leftarrow \ln 16/M_f$
- 6: **end if**
- 7: Compute r_t^{DARE} by (5) with $f_\theta = \kappa g_\theta$
- 8: **for** critic minibatch **do**
- 9: Sample equal numbers of current and replay differentials; update g_θ by one step on (4) with $f_\theta = \kappa g_\theta$
- 10: **end for**
- 11: Update π_φ and V_ω with PPO on r_t^{DARE}
- 12: **end for**

TABLE I

DARE CONFIGURATION SHARED ACROSS ALL SIX SKILLS. THE CALIBRATION RULE IS SHARED; THE RESULTING κ IS DATA-DEPENDENT.

Component	Setting
Differential	single frame, $d = 172$
Semantic factors	$G=7$: root/body pos. (3/45), root/body rot. (6/84), root lin./ang. vel. (3/3), DoF vel. (28)
Direct-sum layer	$m = 146$ per factor (1022 total)
Shared trunk	linear $1022 \rightarrow 512$; output $512 \rightarrow 1$
Activation ρ	ReLU (1-Lipschitz)
Spectral norm.	all linear maps, one power iteration
Normalizer $\mathcal{N}$	per-coordinate mean-abs scale, fixed after 10^8 samples (iter. 382)
Logit calibration	$\kappa = \ln 16/M_f$, once at iter. 382
Critic optimizer	SGD, lr 2.5×10^{-4} , momentum 0.9
Critic minibatch	16384 current + 16384 replay
Replay buffer	2×10^5 differentials
Reward amplitude β	2

whole loop. The same critic architecture, reward definition, and calibration procedure are used across motions; motion-dependent statistics are obtained automatically from the training data.

V. EXPERIMENTS

A. Experimental Setup

All methods are evaluated using the same 28-DoF, 15-body humanoid model adopted in prior physics-based motion-imitation studies. Simulation runs in Isaac Lab at 120Hz, with policy actions applied at 30Hz through PD joint-position control [19], [20]. All four methods are trained under the same simulation environment, training budget, motion clips, random-phase initialization, and evaluation protocol. Each policy is trained for 2000 iterations of 8192 environments $\times$ 32 steps (5.24×10^8 samples), with five independent seeds, and is evaluated on 4096 episodes per seed from randomly sampled reference phases. Actor and value networks are two-layer MLPs (1024, 512), and the PPO settings (clip 0.2, GAE 0.95, $\gamma = 0.99$, SGD with momentum 0.9) are fixed across methods. DARE additionally uses the

normalizer and one-shot logit calibration specified in Table I; the normalizer is fixed after 10^8 samples (iteration 382).

The benchmark contains six skills with substantially different motion durations, contact patterns, and dynamical characteristics: Roll, Climb, Backflip, Crawl, Get-Up, and Spin Kick. Training termination follows each method’s environment configuration (pose termination is enabled for DeepMimic, ADD, and DARE and disabled for AMP). The evaluation harness accumulates each metric over the executed reference-aligned steps and uses the same aggregation rule for every method; terminated trajectories are not padded with synthetic states. Final policies reach essentially the full prescribed evaluation horizon, so early termination has negligible effect on the reported averages. For the macro-averages in Table II, the six skills contribute equally before the five-seed statistics are computed.

We use six complementary measures. The four geometric measures quantify frame-wise imitation fidelity, whereas foot progress and lateral drift capture trajectory-level behavior that is not summarized by instantaneous pose errors. Four describe geometric tracking at two spatial levels: root position and root rotation measure global translation and orientation, while body position and body rotation measure articulated-body tracking. For an evaluation trajectory with B bodies, where r, p_b and q, q_b denote simulated root, body positions and orientations and hats denote reference values, these quantities are

$$\begin{aligned}
 E_{\text{root-pos}} &= \|\hat{r} - r\|_2, & E_{\text{root-rot}} &= \angle(q, \hat{q}), \\
 E_{\text{body-pos}} &= \frac{1}{B} \sum_{b=1}^B \|(\hat{p}_b - \hat{r}) - (p_b - r)\|_2, \\
 E_{\text{body-rot}} &= \frac{1}{B} \sum_{b=1}^B \angle(q_b, \hat{q}_b).
 \end{aligned} \tag{14}$$

where $\angle(\cdot, \cdot)$ is the shortest angle between normalized quaternions. Geometric quantities are evaluated at each reference-aligned timestep and then averaged over executed timesteps and episodes; the two behavior measures are computed per trajectory and then averaged over episodes. The remaining two measures describe motion-level behavior. Let Δr and $\Delta \hat{r}$ be the simulated and reference root displacements over the evaluated trajectory, and let $\Delta p_j, \Delta \hat{p}_j$ be the corresponding displacements of foot j . The reference-projected progress and transverse drift are

$$\begin{aligned}
 \rho_{\text{foot}} &= \min_j \frac{\Delta p_j^\top \Delta \hat{p}_j}{\|\Delta \hat{p}_j\|_2^2}, \\
 \rho_r &= \frac{\Delta r^\top \Delta \hat{r}}{\|\Delta \hat{r}\|_2^2}, \\
 E_{\text{lat}} &= \frac{\|\Delta r - \rho_r \Delta \hat{r}\|_2}{\|\Delta \hat{r}\|_2}.
 \end{aligned} \tag{15}$$

Foot progress is ideal at one, while lateral drift is minimized at zero. All six quantities are averaged over the evaluation trajectories; lower is better except for foot progress, whose target value is one.

TABLE II

MACRO-AVERAGE TRACKING AND MOTION-LEVEL BEHAVIOR UNDER THE MATCHED TRAINING AND EVALUATION PROTOCOL. FOR EACH SEED, EACH METRIC IS MACRO-AVERAGED EQUALLY OVER THE SIX SKILLS; ENTRIES REPORT THE MEAN $\pm$ ONE STANDARD DEVIATION ACROSS FIVE SEEDS. LOWER IS BETTER EXCEPT FOR FOOT PROGRESS, WHICH IS IDEAL AT ONE. DARE USES THE SAME CONFIGURATION ON ALL SKILLS.

(a) Geometric tracking				
Metric $\downarrow$	AMP	DeepMimic	ADD	DARE
Root Pos. [m]	0.9116 $\pm$ 0.084	0.0703 $\pm$ 0.012	0.0725 $\pm$ 0.010	0.0528$\pm$0.008
Body Pos. [m]	0.2169 $\pm$ 0.031	0.0557 $\pm$ 0.009	0.0363 $\pm$ 0.007	0.0313$\pm$0.006
Root Rot. [rad]	0.7311 $\pm$ 0.076	0.1459 $\pm$ 0.021	0.0673 $\pm$ 0.011	0.0529$\pm$0.009
Body Rot. [rad]	0.8098 $\pm$ 0.069	0.2145 $\pm$ 0.028	0.1358 $\pm$ 0.017	0.1246$\pm$0.013
(b) Motion-level behavior				
Metric	AMP	DeepMimic	ADD	DARE
Foot progress $\rightarrow$ 1	1.1335 $\pm$ 0.046	1.0060 $\pm$ 0.018	1.0109 $\pm$ 0.021	0.9989$\pm$0.012
Lateral drift $\downarrow$	0.1127 $\pm$ 0.019	0.0277 $\pm$ 0.007	0.0435 $\pm$ 0.009	0.0171$\pm$0.005

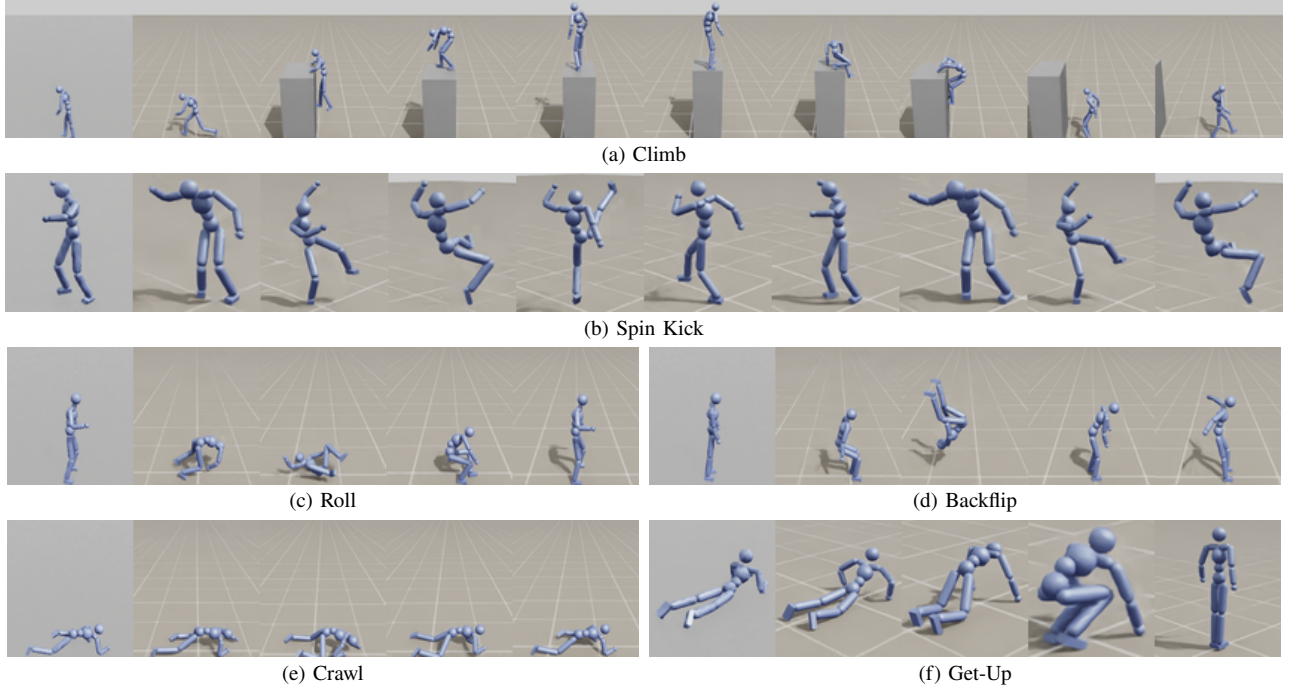


Fig. 1. DARE policies executing the six skills in Isaac Lab using the shared configuration in Table I. Frames are uniformly sampled from each rollout and rendered from a fixed front-view camera. Climb and Spin Kick are shown with ten frames, and the remaining skills with five.

B. Cross-Motion Tracking

Table II asks whether one configuration can serve skills whose contacts, horizons, and velocity profiles differ, and Fig. 1 shows the resulting controllers executing the six skills in Isaac Lab. Using one shared configuration, DARE is evaluated against the three baselines under the matched protocol of Section V-A. Table II reports the macro-average geometric errors and motion-level behavior across the six skills. Using one unchanged configuration across all six skills, DARE obtains the best macro-averaged result on all six reported measures. The improvement is consistent across the two levels of geometric tracking: DARE achieves the lowest root- and body-level position errors as well as the lowest root- and body-level rotation errors. The advantage also extends beyond frame-wise pose fidelity: its foot-progress ratio is closest to the reference value of one, while its normalized

lateral drift is the smallest among all methods. Relative to the strongest baseline for each measure, DARE reduces root-position error by about 25%, root-rotation error by about 21%, and lateral drift by about 38%.

C. Learning Dynamics

Figure 2 compares the five-seed mean body-position and root-rotation errors against environment samples under a matched budget. Roll and Climb provide contrasting motion durations and contact structures. Across both motions, the five-seed mean curves converge to lower body-position and root-rotation errors for DARE under the same sample budget. The learning curves therefore complement the final-checkpoint comparison in Table II.

D. Hardware Deployment

[TODO: Report the robot platform and its degrees of freedom, reference retargeting, domain randomization, observation noise, action filtering, hardware control frequency, and the number of trials and successful executions for each skill. State that the shared DARE reward formulation and calibration rule are used without motion-specific modification.]

VI. CONCLUSION

Manual tracking rewards turn the aggregation of heterogeneous imitation objectives into per-motion engineering. Learned differential rewards remove the component-wise weights, but regularization can still couple the critic’s geometric sensitivity to the score range seen by the policy. DARE treats these as separate design quantities: the factorized critic specifies the geometry, and one closed-form calibration sets the score scale without a motion-specific coefficient. Every objective enters through its own spectrally normalized slot, so feature blocks of unequal dimension enter through equal-width, independently normalized representations, and the score change relative to the unique zero differential is bounded by the residual; the critic is then rescaled once, in closed form, so that the anchor and the typical residual sit at the two ends of the reward slope’s transition. Across the six skills, DARE attains the lowest macro-average for all four geometric errors, the foot-progress ratio closest to one, and

the smallest lateral drift in Table II. The formulation also provides a direct path toward multi-clip training and other morphologies, where additional objectives can be introduced as semantic blocks.

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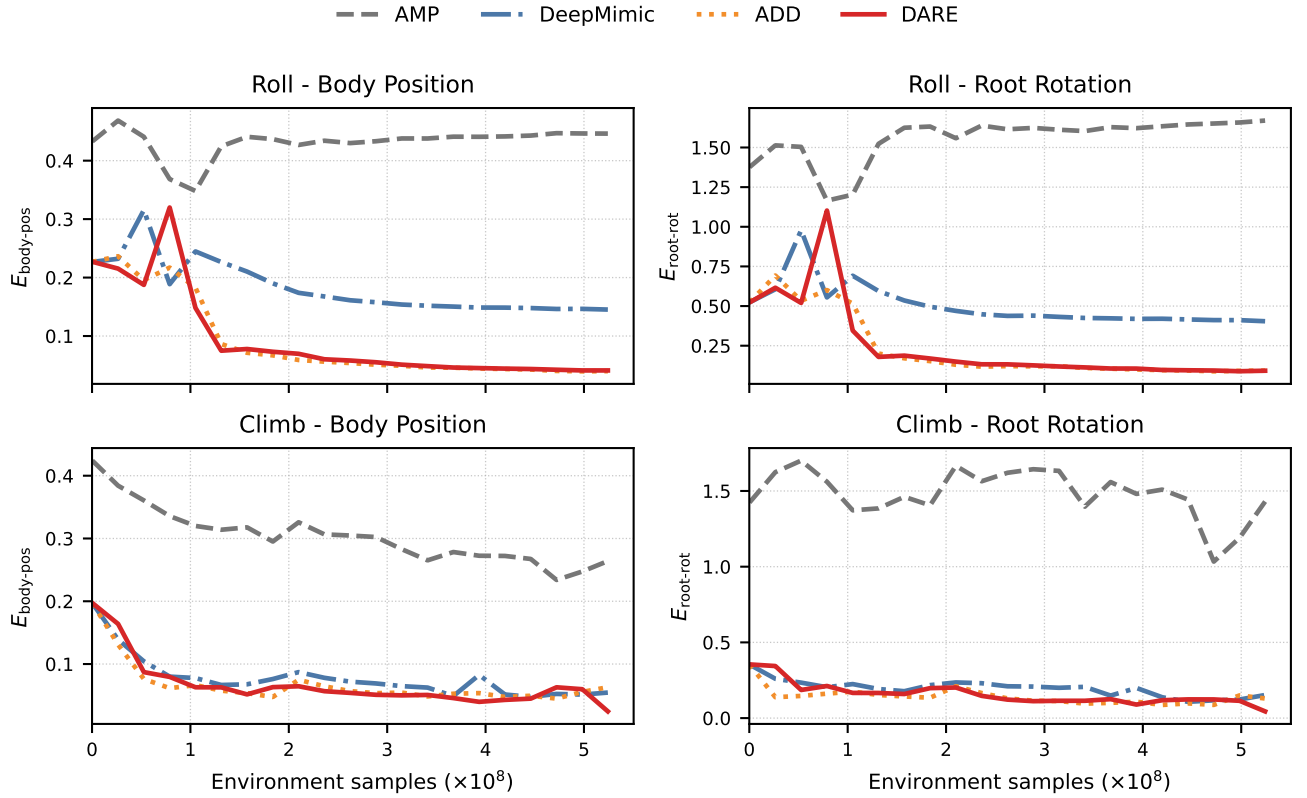


Fig. 2. Learning curves for Roll and Climb under the matched training budget. Curves show the mean over five independent seeds. The four panels report body-position and root-rotation errors versus environment samples, representing articulated translational fidelity and global orientation tracking, respectively. Lower is better.

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